

CE 310

Sensitivity Analysis

Week 12 · Session 1 · How Uncertainty in Inputs Propagates to Outputs

University of Arizona · Dr. Wooyoung Jung · Fall 2026

\$60,000 Left in the Investigation Budget

90 seconds · you can buy exactly one

A highway embankment slope computes to **FS = 1.31** against a required **1.50**. Buttrressing it costs \$2.1M. You get one more investigation before that decision.

Uncertain input	Best estimate	Plausible range
Friction angle ϕ'	31°	29° – 33°
Groundwater, as fraction of slope depth	0.40	0.15 – 0.85
Soil unit weight γ	125 pcf	122 – 128 pcf

\$60,000 buys six borings with direct shear, *or* a piezometer array read through one wet season, *or* a full unit-weight in-class exercise program. **Which one do you buy?**

✓ Buy the Piezometers

The input nobody measured is the one that decides the slope

Move each input across its plausible range, holding the others at their best estimate:

Input moved across its range	Resulting FS	Swing
Groundwater	0.97 – 1.49	0.52
Friction angle ϕ'	1.21 – 1.40	0.19
Soil unit weight γ	1.30 – 1.31	0.01

At the wet end of its range, **FS = 0.97 — the slope has already failed.** No borehole result for ϕ' can move the answer that far, and γ cannot move it at all.

💡 What ranks an input is not how strongly the output responds to it, but that **times how wrong the input could be.** Rank by the product — never by which parameter feels most important.

Where a Range Comes From

The ranking multiplies response by uncertainty. The response falls out of the model — **the range is the part you have to go and find**, and the places you can find one are not equally defensible:

Source of a range	Example	How well it holds up
A published tolerance	survey class, ASTM gradation limits	strongest — someone else already argued it
Historical spread	what this item actually cost on 555 past jobs	strong, if those jobs resemble this one
A model's own residuals	± 1 RMSE around a fitted price	honest, and usually the widest
Expert judgement	" ϕ' is probably 29° to 33° "	weakest — and the most common

💡 Groundwater won the slope problem on the **fourth** row. Nobody had measured it, so its range was a judgement — wide, soft, and decisive.

Which Range Would You Defend?

90 seconds · one of these will not survive the hearing

You are pricing a bid item and you write down four ranges. The owner's reviewer asks where each one came from.

Input	Range you wrote	Where it came from
A Concrete strength f'_c	4,000 – 4,800 psi	the mix design submittal
B Unit price	$\times 0.57 - \times 1.75$	residual spread of a fit on 555 jobs
C Quantity	$\pm 15\%$	"that's what take-off disputes usually run"
D Escalation	2% – 8%	the last five years of the index

Which range is weakest — and does weak mean too narrow, or too wide?

✓ C — and Weak Almost Always Means Too Narrow

The soft number is the one with a story instead of a source

C is the only range with no record behind it. A, B and D each point at a document you can hand the reviewer. C points at a habit.

⚠ The error runs one way. Nobody ever guesses a range that is **too wide** — a wide guess feels like confessing ignorance, so guesses come in tight. A soft $\pm 15\%$ that is really $\pm 30\%$ halves quantity's bar and drops it down a ranking it should have led.

💡 Every bar on a tornado is a claim about a range. If you cannot say where the range came from, the bar is decoration — and the input most likely to have been guessed is the one nobody measured.

The Same Question, in Dollars

The slope problem was a factor of safety. Swap it for a number you will meet far more often — **what a bid item costs:**

$$\text{item cost} = Q \times P \times (1 + \text{overhead}) \times (1 + \text{escalation})$$

Four inputs, none of them known exactly:

Input	Where the uncertainty comes from
Quantity Q	a take-off dispute — plans and field measure disagree
Unit price P	what contractors actually charge varies job to job
Overhead	firms carry different indirect structures
Escalation	depends on how long the contract runs

The question is the same: which one decides the number you sign?

Two Jobs, One Method

Civil — an agency programme

A district lets hundreds of contracts a year. The question is not "which input dominates this job" but where a fixed improvement budget does the most good across all of them.

Same tornado, aggregated.

Architectural — a building contract

A completion date carries liquidated damages. Duration is forecast from contract value, and the forecast is uncertain.

Same tornado, in days rather than dollars.

💡 The data on these slides is highway work, because public agencies publish it and private building owners do not. The **method** is neither civil nor architectural — it belongs to anyone who signs a number they are not certain of.

For a Fitted Model, the Coefficient IS the Sensitivity

Week 11 fitted the price of roadway excavation against job size:

$$\log_{10} P = \beta_0 + \beta_1 \log_{10} Q + \beta_2 \cdot \text{LOW}$$

Sensitivity to quantity = β_1 . There is nothing to re-run — the number came out of the regression.

On a log–log fit β_1 is an **elasticity**: the percentage change in price per one percent change in quantity.

A tenfold job raises $\log_{10} Q$ by exactly 1, so price is multiplied by $10^{\beta_1} = 10^{-0.166} = 0.682$.

That is a **31.8% lower** unit price. For a large change you *exponentiate*; multiplying β_1 by a percentage only works while the change is small.

Why this matters: the ranking you are about to build costs nothing to compute for any input that enters a fitted model. The work is in the *ranges*, not the responses.

Applied to Bid Prices

At a design quantity of **2,000 CY** — a routine job, near the median:

	At 2,000 CY
Predicted unit price	\$21.26 per CY (winning bid)
Elasticity	$\beta_1 = -0.166$
In dollars	$dP/dQ = -\$0.0041$ per CY, per additional CY

Note the sign. **More work, lower unit price** — mobilisation and setup spread over more cubic yards.

⚠ A small elasticity does **not** mean quantity is unimportant. It means quantity barely moves the price *per CY*. Cost is quantity $\times$ price, so quantity still enters directly — and the tornado will show it second.

One-at-a-Time (OAT) Perturbation

The coefficient is exact for an *infinitesimal* change. Engineers work with real ones, so check it numerically: **move one input, hold everything else fixed, measure what happens.**

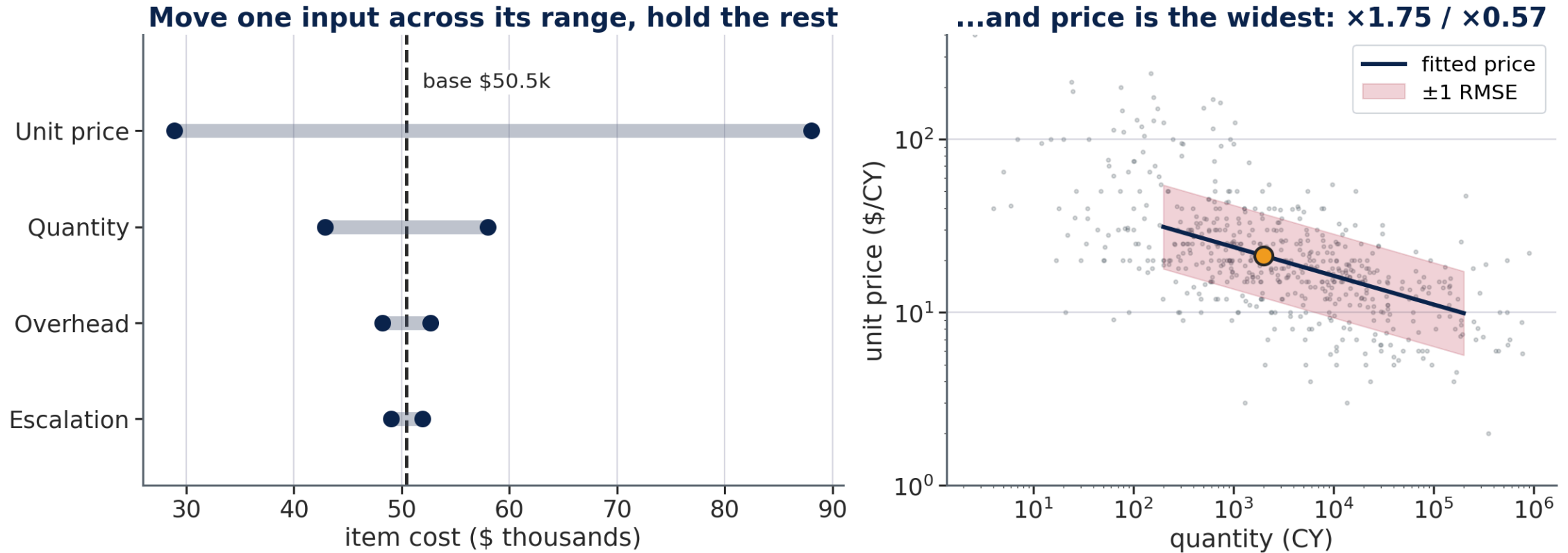
Move quantity **+15%** — a routine take-off dispute:

Quantity	Predicted unit price
2,000 CY	\$21.2636
2,300 CY (+15%)	\$20.7761
	-2.29%

The elasticity predicted $15 \times -0.166 = \mathbf{-2.49\%}$. Close, not equal — and the gap is the curvature that a straight-line reading of the slope ignores.

💡 That gap belongs to *this* model, not to coefficients in general. Where a model is **linear in the predictor itself**, the coefficient is the sensitivity exactly, at every value of that predictor. Curvature appears here only because the fit is in logs.

What OAT Actually Does



Left: each input swept across its own plausible range while the other three hold at base. Right: where the widest of those ranges comes from — it is **measured**, not assumed. The band is the model's own residual spread.

Scaling Up — The Programme Aggregate

One job's exposure is not what an agency budgets against. Across every **awarded** excavation contract in the data:

Awarded excavation work	
Contracts	555
Programme quantity	16,486,177 CY
Programme cost	\$203,833,501
If every unit price were 1 RMSE high	+\$151,885,987 (74.5%)

⚠ Aggregate over **awarded work only**. Every project appears in a bid tabulation once per bidder, so summing across all rows counts each job several times — here, 6.7× the truth.

Quick Check — Which Input Dominates?

90 seconds · rank them

Base item cost at 2,000 CY: **\$50,459**. Each input has its own plausible range:

Input	Low	Base	High
A Quantity	−15%	2,000 CY	+15%
B Unit price	×0.57	\$21.26/CY	×1.75
C Overhead	8%	13%	18%
D Escalation	2%	5%	8%

Which produces the largest swing in item cost — and is it because cost responds most strongly to it?

✓ Unit Price — and Not Because Cost Responds to It

Input	Swing	% of base
Unit price	\$59,144	117.2%
Quantity	\$15,138	30.0%
Overhead	\$4,465	8.8%
Escalation	\$2,883	5.7%

Cost is **linear in all four**. None has a steeper response than any other, so response cannot be what separates them. What separates them is the **width of the range** — and price's range is the model's own residual spread, $\times 0.57$ to $\times 1.75$.

💡 Rank by **response $\times$ uncertainty**. When the responses are equal, the ranking is pure uncertainty — which is the honest version of what the slope problem showed with groundwater.

One Swing, Computed by Hand

The last slide handed you four swings. Here is where one of them came from — **the overhead swing**, start to finish. The cost model is the one from earlier:

$$\text{item cost} = Q \times P \times (1 + \text{OH}) \times (1 + \text{ESC})$$

Step	Overhead at 8%	Overhead at 18%
$Q \times P = 2,000 \times \21.2636	\$42,527	\$42,527
$\times (1 + \text{OH})$	$\times 1.08 \rightarrow \$45,929$	$\times 1.18 \rightarrow \$50,182$
$\times (1 + \text{ESC}), \text{ i.e. } \times 1.05$	\$48,226	\$52,691

Swing = \$52,691 – \$48,226 = \$4,465. Quantity and price never moved. That is what *one-at-a-time* means, in arithmetic.

💡 Every bar on that tornado is this calculation, run twice and subtracted. Nothing more sophisticated is happening anywhere in the method.

Now Do One Yourself — the Escalation Swing

90 seconds · same method, one input over

Escalation runs **2% to 8%**, base 5%. Everything else holds at base: **$Q = 2,000 \text{ CY} \cdot P = \$21.2636/\text{CY} \cdot \text{overhead} = 13\%$** .

The first two steps are already done for you, because neither of them moves:

Step	Value
$Q \times P$	\$42,527
$\times (1 + \text{OH}), \text{ i.e. } \times 1.13$	\$48,056

Finish it. What is the escalation swing — and will it be bigger or smaller than overhead's \$4,465?

✓ \$2,883 — the Shortest Bar on the Chart

And it is short because the range is narrow, not because cost ignores it

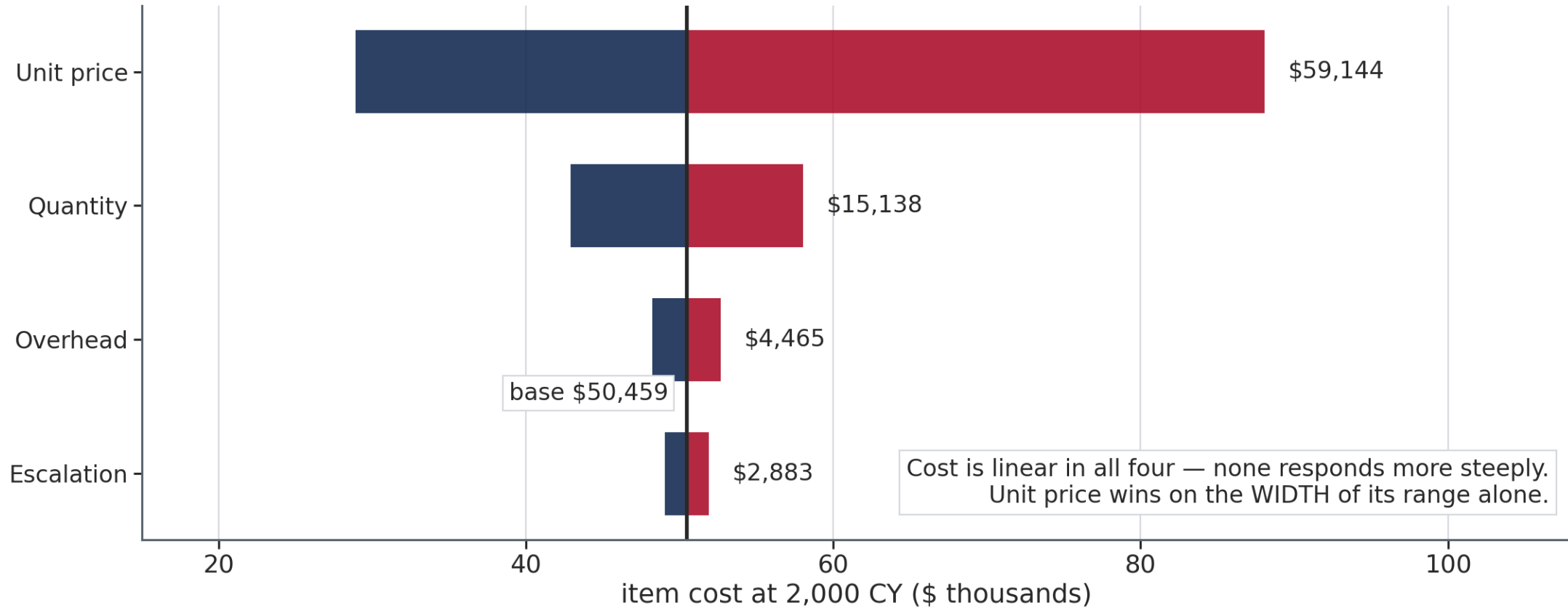
Step	Escalation at 2%	Escalation at 8%
Running total	\$48,056	\$48,056
× (1 + ESC)	× 1.02 → \$49,017	× 1.08 → \$51,900

Swing = \$51,900 – \$49,017 = \$2,883.

Cost responds to escalation exactly as readily as it responds to overhead — both enter the model the same way. Overhead's band is **8 to 18 points wide**; escalation's is **2 to 8**. The wider band produced the bigger bar.

💡 Four runs of this arithmetic and the tornado is finished. The judgement was never in the multiplying — it was in choosing 2% to 8% rather than 1% to 15%.

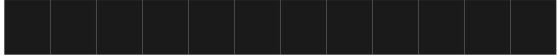
Reading a Tornado Diagram



Bars sorted widest first, all at the same design point. The name comes from the shape, and the shape is the deliverable.

Back to the Slope — the Same Chart, in Factor of Safety

The bid item is not a special case. The embankment from the opening slide has a tornado too, and you have already computed it:

Input moved across its range	Swing in FS	
Groundwater	0.52	
Friction angle ϕ'	0.19	
Soil unit weight γ	0.01	

Same construction, same sort, same reading. Only the units changed — dollars became a factor of safety, and *buy the piezometers* became *buy the quotes*.

💡 Nothing in the method knows what it is ranking. Any model with uncertain inputs and one output you care about has a tornado behind it — a cost, a settlement, a peak demand, a completion date.

Does the Ranking Depend on Who Is Bidding?

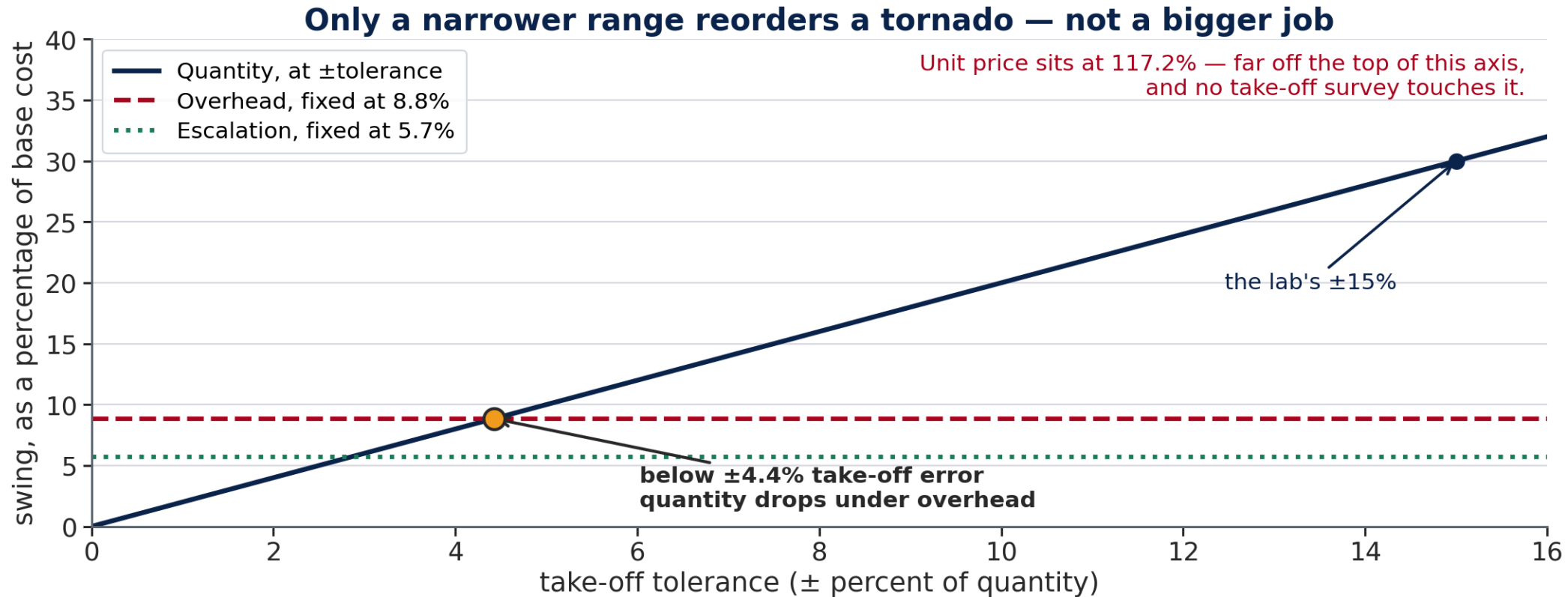
The additive model gives winners and losers one shared slope. The **interaction model** lets the slope itself differ:

At 2,000 CY	Predicted unit price
Winning bid	\$21.45
Losing bid	\$24.07
Ratio	1.122

Losing bids run about **12% higher** at this job size — and the gap is not constant, because the two lines have different slopes.

For estimating, the relevant model is the winning one. You are not forecasting what someone will bid; you are forecasting what the job will cost.

What Would Move the Ranking



A tighter take-off shrinks quantity's swing and can push it below overhead. A **bigger job cannot** — every swing scales together, so the ranking is untouched.

A Tornado Is Computed at One Point

Every bar so far was measured at **2,000 CY**, with the other three inputs at base. A tornado describes **one design point**, not the model everywhere.

Here, moving the design point rescales every bar together:

Design point	Quantity swing	Overhead swing	Order
500 CY	smaller	smaller	unchanged
2,000 CY	\$15,138	\$4,465	unchanged
20,000 CY	larger	larger	unchanged

The order holds because this cost model is **linear in every input**.

⚠ Do not carry that guarantee to the next model. Put a threshold in it — a slope fine at 30° and gone at 33° — and an input can sit at the bottom of the tornado at one design point and near the top a little further along. When the model has a cliff in it, the ranking holds only where you computed it.

How Accurate Does the Take-off Need to Be?

Sensitivity analysis converts into a **specification** — the useful output for anyone writing a scope of work.

Find the quantity tolerance at which the take-off swing falls below the overhead swing, the point past which more survey effort stops changing anything:

At 2,000 CY	
Overhead swing	\$4,465
Take-off tolerance that matches it	±88.5 CY (±4.4% of 2,000 CY)

Past that point, buying a tighter take-off is buying accuracy in the wrong input.

Quick Check — Is the Survey Worth Buying?

Work it out · 60 seconds

A detailed quantity survey narrows the take-off range from **$\pm 15\%$** to **$\pm 3\%$** .

Current swings at 2,000 CY: unit price **\$59,144** · quantity **\$15,138** · overhead **\$4,465** · escalation **\$2,883**

What is the quantity swing after the survey — and does the ranking change?

- **A)** $\approx$ \$3,000 — quantity drops to third, behind overhead
- **B)** $\approx$ \$3,000 — quantity stays second
- **C)** $\approx$ \$12,000 — quantity stays second

✓ A — And It Reordered the Wrong End of the List

Input	Before ($\pm 15\%$)	After ($\pm 3\%$)
Unit price	\$59,144	\$59,144 — untouched
Quantity	\$15,138	\$3,028
Overhead	\$4,465	\$4,465
Escalation	\$2,883	\$2,883

Quantity falls from second to third. Unit price still leads by **more than an order of magnitude**.

💡 A sensitivity ranking tells you where to spend, but it also tells you where spending is *wasted*.
Buying accuracy in an input that was never going to decide the answer is the most common way engineering money is spent on nothing.

One Improvement, 555 Contracts

90 seconds · you can fund exactly one

The district lets **555 excavation contracts** a year — **\$203.8M** of awarded work. You can fund one programme-wide improvement for next season:

	Improvement	What it narrows
A	Require a certified take-off on every job over \$100k	the quantity range
B	Build a district price database from three years of tabs	the price range
C	Negotiate a standard overhead rate with local contractors	overhead

Which buys the most — and does the programme answer differ from the single-job answer?

✓ B — and the Single-Job Answer Carried Over

Which is worth checking, not assuming

Improvement	Single-job swing	Programme scale †
B Price database	\$59,144	~\$33M
A Certified take-off	\$15,138	~\$8M
C Overhead agreement	\$4,465	~\$2M

⚠ † Order-of-magnitude only — the 2,000 CY swings scaled by 555 jobs. The *ranking* is the reliable part; the totals are an illustration.

Every swing scales by the same factor, so the order survives — the same reason a **bigger job** could not reorder the tornado.

💡 Worth checking, not assuming. A ranking that held for one job becomes programme policy only when the cost model is linear and the jobs resemble each other — break either and you must re-rank.

Limitations of OAT Analysis

Everything so far moved **one input at a time**. That has four costs:

- **No simultaneity.** OAT never tests quantity high *and* price high together, which is the case that actually breaks a bid.
- **No probability.** A swing is a range, not a likelihood. Nothing here says how *likely* the high corner is.
- **Only what the model contains.** You can perturb an input only if it appears in the equation. Anything the model leaves out — a labour dispute, a differing site condition — has no bar on the chart, however much it drives the real cost.
- **Corner stacking is worse.** Push all four to their worst end at once and this item spans **170.3%** of base cost — a number no estimator would budget against, because all four going wrong together is far less likely than any one of them doing so.

The Bar That Is Not on the Chart

⚠ A tornado ranks **the inputs the model contains**. Anything the equation never mentions has no bar at all — and on a chart, a missing bar looks exactly like a short one.

For this cost model, the ranking is silent on:

- **A differing site condition** — rock where the borings showed soil
- **A subcontractor default**, or a labour dispute
- **Weather** shutting the site down for three weeks
- **A design change** after award

Any one of these can cost more than the entire **\$59,144** price swing that led the ranking.

💡 The honest limit of the method: it ranks the risks *you thought to model*, and is silent on the ones you did not — which are usually why projects actually go wrong.

What Monte Carlo Adds

Week 13 samples all four inputs from distributions at once, runs the cost model thousands of times, and reports the distribution of results.

The output stops being *"if price is one RMSE high, cost rises \$59,144"* and becomes *"there is a 90% chance this item lands between \$38,000 and \$76,000."*

That second sentence is the one you can put in a contingency decision.

Bridge to Machine Learning — Feature Importance

Sensitivity concept	ML equivalent	Connection
Coefficient = analytical sensitivity	Feature importance (linear models)	Coefficient $\times$ input range
OAT perturbation	Partial dependence plot (PDP)	Change one feature, hold others
Tornado diagram	Feature importance ranking	Which input drives the output most?
Range width, not response	Permutation importance	Shuffle a feature; how much is lost?

Feature importance is the same question this lecture asked: which input drives the output most? SHAP (SHapley Additive exPlanations) is the modern tornado diagram — it decomposes a single prediction into contributions from each feature.

Three Deliverables This Week

⚠️ **No Wednesday tutorial this week.** Veterans Day cancels Wed Nov 11 — the tutorial is posted as a **recording** and you take the quiz on your own time. Friday runs the exercise as one full session.

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Nov 17
In-Class Exercise	The Colab notebook — all sections, plots, and written answers	Friday, one session · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

Office hours run as normal.

💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Nov 17; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 13: Monte Carlo Simulation

Sensitivity analysis answered: *if one input shifts by X , the cost shifts by Y .*

Monte Carlo answers: *if every input has a distribution, what is the distribution of the cost?*

The method: instead of moving quantity to $\pm 15\%$ as one scenario, draw 10,000 quantities from a realistic distribution — and 10,000 prices, and overheads, and escalations — run the cost model for each, and histogram the results.

The output is a **prediction interval**: "there is a 90% chance this item lands between \$38,000 and \$76,000." That is what a contingency percentage is actually set from, and it is what B6's 170.3% corner could never give you.